Mohit Saroha

Lead Architect (NLP & GenAI)

Email: w3mohit@gmail.com

Contact: 8826155190

PROFESSIONAL OUTLOOK / SUMMARY

- Lead Architect with **13+ years of IT experience**, including **6+ years in Conversational AI & Chatbots** and **3+ years in Generative AI (LLMs, RAG, Agentic AI)**. Strong expertise in **Python-centric development** with hands-on experience across **AWS, GCP, and Azure**.
- Possesses a solid foundation in **Data Structures and Algorithms**, with open-source contributions to **Narasimha Karumanchi (Career Monk) – Python DSA repository**, and additional advanced problem-solving work maintained in private repositories. Actively committed to **community knowledge sharing and mentoring**.
- Experienced in **system design and architecture**, having designed and delivered scalable platforms using **microservices, microfrontends, and event-driven architectures**, optimized for **high availability, low latency, and cloud-native deployment**.

HAND-ON-TECHNOLOGIES

⚙️ **Domains:** Insurance, Cybersecurity, Automotive, Conversational AI, E-Commerce Platforms, Big Data Systems, Enterprise Resource Planning (ERP)

🧠 **AI / GenAI:** Generative AI (GenAI), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Agentic AI, LangChain, LangGraph, Hugging Face, OpenAI APIs, Whisper (STT), ElevenLabs (TTS)

🖥️ **Languages:** Python, JavaScript, TypeScript, SQL, Go, C#, Node.js

⚙️ **System Design:** Microservices Architecture, Microfrontends, Event-Driven Architecture, Distributed Systems, Scalable System Design

🎨 **Frontend:** React 18, Angular, NgRx, TypeScript, Progressive Web Apps (PWA)

🛠️ **Backend:** FastAPI, Django, Flask, Express.js, Node.js, RESTful APIs, Async Processing

☁️ **Cloud & DevOps:** Google Cloud Platform (GCP), Amazon Web Services (AWS), Microsoft Azure, Docker, CI/CD Pipelines, Nginx

🎙️ **Voice & Video:** Text-to-Speech (TTS), Speech-to-Text (STT), Voice Cloning, Real-Time Streaming, Server-Sent Events (SSE)

🔐 **Auth & Security:** JWT Authentication, OAuth 2.0, Azure Active Directory (Azure AD), API Gateway, Role-Based Access Control (RBAC)

🧪 **Testing:** Jest, Playwright, Locust (Performance Testing), ELK Stack (Logging & Monitoring)

📊 **Databases:** PostgreSQL, MongoDB, Redis, Snowflake, Vector Databases, FAISS, Azure Cognitive Search, Azure Cosmos DB, Elasticsearch

👥 **Leadership:** Team Leadership, Technical Mentoring, Agile/Scrum, Stakeholder Management, Architecture Reviews

🔍 **Big Data:** Pandas, NumPy

🤖 **Chatbots:** RASA, MindMeld, Conversational Design, AI Bot Architecture, Dialogue Flow Design, LLM-powered Chatbots

📡 **Streaming:** Server-Sent Events (SSE), Django Event Stream, Real-Time Data Streaming, Voice Streaming, Video Streaming

CAREER DETAILS

- **Technical Architect –** Generative AI & Platform Engineering @ Fractal Analytics Pvt. Ltd. | Apr 2023 – Present
- **Team Lead** @ Capgemini India Pvt. Ltd. | Jun 2021 – Mar 2023
- **Senior Technical Associate @** Policybazaar | Jan 2019 – Mar 2021
- **Lead Technical Architect** R S S R Solutions Pvt. Ltd. (YantraWorld – yantraworld.in) | May 2012 – Jan 2019

ACADEMIC QUALIFICATIONS

- **B.Tech** CSE — 2013 | 72%, XII (CBSE) – 75%, X (CBSE)– 83%

PROJECTS DETAILS **(HIGH IMPACT ONLY)**

- **Avatars**:- Build Your Own Bot/Avatar. This is a **Chat Bot(Text+voice) As a digital Replica of Any body who uploads their documents** at avatars.fractal.ai

    - Built **BYOA (Build Your Own Avatar)** platform enabling leaders and coaches globally to create AI-powered avatars using **OpenAI APIs** with advanced conversational intelligence.

    - Designed an **Agentic AI architecture in Python**, enabling multi-step reasoning, memory chaining, tool usage, and contextual awareness across sessions.

    - Developed scalable backend microservices using **FastAPI**, deployed on **Azure Virtual Machines** and **Azure App Service** for high availability.

    - Containerized services with **Docker** and managed orchestration and scaling using **Azure Kubernetes Service (AKS)** to handle production traffic spikes.

    - Implemented **Azure Cognitive Search (Vector Search)** with **Azure Cosmos DB (Vector Database)** to support semantic search and Retrieval-Augmented Generation (RAG) for intelligent knowledge retrieval.

    - Stored structured data (users, bot configs, conversation history) in **Azure SQL Database**, and media files (audio, voice samples) in **Azure Blob Storage**.

    - Integrated **Whisper** for speech-to-text and **ElevenLabs** for real-time cloned voice streaming, delivering personalized audio responses.

    - Used **Redis** for caching, session management, and fast memory retrieval to reduce latency in real-time conversations.

    - Implemented **Celery with Redis as a message broker** for handling background tasks such as audio processing, embedding generation, and asynchronous AI workflows.
    - Ensured performance and scalability for 1,000–3,000 concurrent users using **Azure Load Balancer, Azure CDN, Azure Auto Scaling, Azure**

Monitor, Application Insights, and Azure DevOps for monitoring and CI/CD automation.
-

- **MarshallGoldsmith.ai**:- Marshall Goldsmith is Worlds No. 1 Leadership coach (wiki)This is a **Chat Bot(Text+voice) As a digital Replica of Marshall Goldsmith**
  - All he knows is available to all for free.
  - Architected MG-BOT as a scalable RAG (Retrieval-Augmented Generation) system, combining LLM reasoning with vector-based semantic search over Marshall Goldsmith's proprietary knowledge base.

  - Built a distributed microservices architecture using Django (Python) and ExpressJS (TypeScript), separating AI orchestration from structured data and database services.

  - Implemented a vector embedding pipeline on GCP to generate and store embeddings for semantic similarity search and contextual knowledge retrieval.

  - Deployed containerized services using Docker on Google App Engine with auto-scaling, ensuring stateless, horizontally scalable infrastructure.

  - Designed real-time token streaming from LLM → Django backend → React frontend using Django EventStream and Pushpin (SSE gateway) for scalable live responses.

  - Integrated Whisper (speech-to-text) and ElevenLabs (voice cloning) to enable real-time AI conversations in Marshall's cloned streaming voice.

  - Used Celery (without Redis) for distributed asynchronous task processing, handling email campaigns, embedding generation, indexing, and audio processing.

  - Configured Nginx as a reverse proxy and traffic router to manage high concurrency and optimize request handling.

  - Conducted stress and load testing using Locust to validate performance under enterprise-level traffic conditions.

  - Engineered the system to be cloud-native, fault-tolerant, and fully horizontally scalable on GCP, with independent scaling of RAG retrieval, streaming services, Celery workers, and voice processing components.

- **CNB-BOT**:- This is a **Chat Bot for Enterprise networking Migration**
  - CNB-BOT is being used by the Employee and partner of Cisco.
  - Customer Website: **cnb.cisco.com**

- **This is the EN bot migrated in RASA**
- It is built on **Python, Django, LLMs, RASA, Openshift, Docker, Snowflake, unittest, Locust, Redis, Nginx, ElasticSearch, ELK FileBeat, Cisco WebEx. React**

- **Cyber Security Suite – Vulnerability Scanner & Phishing Detection Dashboard**
  - 🧩 **Architecture Overview**
  - Frontend: React + Tailwind (PWA-ready)
  - Vulnerability Scanner Service → Python FastAPI
  - Phishing Detection Service → Python FastAPI + ML
  - Auth/User/Role Management → Node.js + Express
  - Async Jobs: Redis Queue + Celery
  - Database: MSSQL + Elasticsearch (for logs), MongoDB (initial scan cache)
  - Deployment: Docker + Nginx + PM2
  - Authentication: JWT-based Role-based access control
  - DevOps Tools: GitHub Actions + ELK Stack for monitoring
  - ***Duration & MyRole:***- 1.5years~ *Architectured and developed its frontend/backend, gathering requirements from the product team and delivering them.*

- **PB-ERP-Revenue-Management-System**:-
  - This is a module of PBERP which will be used to automate the manual process of RMS which includes Revenue Generation, Invoice Making, Payment collection, Contracts Management.
  - It works like doing all these things under contractual conditions.
    Domain is **http://pberp.policybazaar.com**
  - It is built on Python Django, Node.JS, TypeScript, REST, Angular-7, SQL Server.
  - Two members are handling it whole from the backend to the frontend.
  - ***Duration & MyRole:***- 2years~ *Architectured and developed its frontend as structured as PBERP-PO, gathering requirements from the product team and delivering them.*

- **Yantra World**:- This is a **B2B E-Commerce Complete Solution.**
  - FrontEnd is in Angular. And backend is in NODE.JS, TypeScript Python Django
  - Deployed on digital ocean, AWS S3, AWS RDS, NGINX
  - Contains:
    - **Ledger App** for balance amount maintenance per user.
    - **Inventory Managemen**t of warehouse.
    - SalesMan's Sales **Bag management** system.
    - Integrated Whatsapp Apis for communication.
    - Attendance.

    - Sales Audit per salesman.
    - Attendance management is WIP.
    - Salesman Wallet is in WIP.
    - Custom User and Salesman Analytics for product interaction.
    - Push notification.
    - Progressive web app(PWAs).
- Wholly developed, host, deployed and managed by me.
- Customer Website: **yantraworld.in**
- ***Duration & MyRole*****:** *6Years ~ I designed this app end to end, from UX to DB to Domain and resources management.*

DECLARATION

I hereby declare that the information furnished above is true to the best of my knowledge and belief.

(Mohit Saroha)